

Power System Study Guide – Extended Edition

From Classical Network Analysis to Inverter-Based Resources, Grid-Forming Control, and
Microgrid Fault Ride-Through

Generated Study Notes

July 13, 2026

Abstract

This extended guide summarizes core and advanced power-system topics: per-unit normalization, bus modeling, power-flow equations, voltage and angle stability, faults, protection, synchronous-machine dynamics, inverter-based resources (IBRs), grid-following (GFL) and grid-forming (GFM) controls, the stability impact of high IBR penetration, and practical ride-through concepts for microgrids during grid faults. Additional worked details, design checklists, modeling assumptions, and control/protection notes are included to make the material useful as both a review sheet and an implementation reference.

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1 Big Picture

An electric power system converts mechanical, chemical, solar, wind, or stored energy into electric power, transports it through transmission and distribution networks, and delivers it to loads with acceptable voltage, frequency, reliability, power quality, and safety. The traditional grid is dominated by synchronous generators. Modern grids increasingly include inverter-based resources such as wind, solar photovoltaic (PV), battery energy storage systems (BESS), and power-electronic loads.

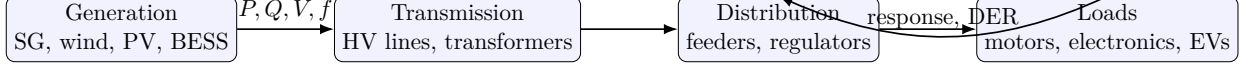


Figure 1: A power system is a controlled energy-conversion network.

2 Phasors, Complex Power, and Per-Unit

2.1 Sinusoidal Steady State

For a sinusoidal voltage,

$$v(t) = \sqrt{2}V_{\text{rms}} \cos(\omega t + \theta), \quad (1)$$

the phasor is

$$\tilde{V} = V_{\text{rms}} \angle \theta. \quad (2)$$

For balanced three-phase systems, a single-phase equivalent is commonly used. Line-to-line and phase quantities obey

$$V_{LL} = \sqrt{3}V_{\phi}, \quad S_{3\phi} = \sqrt{3}V_{LL}I_L. \quad (3)$$

2.2 Complex Power

Complex power is

$$\tilde{S} = P + jQ = \tilde{V}\tilde{I}^*, \quad (4)$$

where P is real power and Q is reactive power. The power factor is

$$\text{pf} = \cos \phi = \frac{P}{|\tilde{S}|}. \quad (5)$$

For inductive loads, current lags voltage and $Q > 0$ under the usual load sign convention.

2.3 Per-Unit System

Per-unit quantities reduce transformer-ratio bookkeeping and put impedances on a comparable scale:

$$X_{\text{pu}} = \frac{X_{\text{actual}}}{X_{\text{base}}}. \quad (6)$$

For a three-phase base,

$$Z_{\text{base}} = \frac{V_{LL,\text{base}}^2}{S_{3\phi,\text{base}}}, \quad I_{\text{base}} = \frac{S_{3\phi,\text{base}}}{\sqrt{3}V_{LL,\text{base}}}. \quad (7)$$

Changing base:

$$Z_{\text{pu,new}} = Z_{\text{pu,old}} \left(\frac{S_{\text{base,new}}}{S_{\text{base,old}}} \right) \left(\frac{V_{\text{base,old}}}{V_{\text{base,new}}} \right)^2. \quad (8)$$

3 Network Modeling

3.1 Transmission-Line Models

For a short line, shunt capacitance can often be neglected:

$$\tilde{V}_S = \tilde{V}_R + \tilde{I}(R + jX). \quad (9)$$

For medium and long lines, the nominal- π model includes shunt admittance $Y/2$ at both ends.

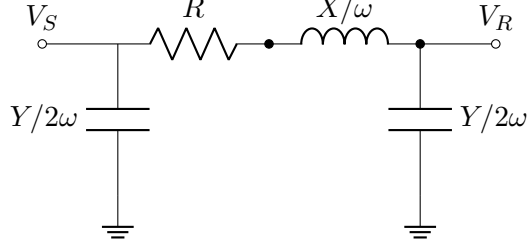


Figure 2: Nominal- π transmission-line model.

3.2 Admittance Matrix

Nodal current injection satisfies

$$\mathbf{I} = \mathbf{Y}_{\text{bus}} \mathbf{V}. \quad (10)$$

For a branch between buses i and k with series admittance y_{ik} :

$$Y_{ii} \leftarrow Y_{ii} + y_{ik} + y_{\text{sh},i}, \quad (11)$$

$$Y_{kk} \leftarrow Y_{kk} + y_{ik} + y_{\text{sh},k}, \quad (12)$$

$$Y_{ik} = Y_{ki} \leftarrow -y_{ik}. \quad (13)$$

Transformer off-nominal tap ratios and phase shifters modify these entries and must be modeled carefully in power-flow and short-circuit studies.

4 Bus Types: Slack, PV, and PQ

Power-flow analysis classifies buses by which variables are specified and which are solved.

Bus type	Known quantities	Unknown quantities	Typical device
Slack / swing	$ V , \delta$	P, Q	Reference generator
PV / generator	$P, V $	Q, δ	Generator with AVR
PQ / load	P, Q	$ V , \delta$	Load or fixed-power DER

Table 1: Power-flow bus types.

- **Slack bus:** supplies the real and reactive mismatch after all other injections and losses are determined. It sets the system angle reference.
- **PV bus:** real power and voltage magnitude are specified. Reactive output is solved, but if it violates limits, the bus is typically converted to PQ with Q fixed at the limit.
- **PQ bus:** real and reactive powers are specified. Voltage magnitude and angle are solved.

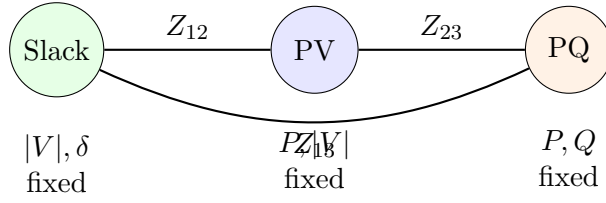


Figure 3: Three classical power-flow bus types.

5 AC Power Flow

5.1 Power Injection Equations

Let $V_i = |V_i| \angle \delta_i$ and $Y_{ik} = G_{ik} + jB_{ik}$. The complex power injected at bus i is

$$S_i = P_i + jQ_i = V_i I_i^* = V_i \left(\sum_{k=1}^n Y_{ik} V_k \right)^* . \quad (14)$$

The real and reactive power equations are

$$P_i = |V_i| \sum_{k=1}^n |V_k| (G_{ik} \cos \delta_{ik} + B_{ik} \sin \delta_{ik}) , \quad (15)$$

$$Q_i = |V_i| \sum_{k=1}^n |V_k| (G_{ik} \sin \delta_{ik} - B_{ik} \cos \delta_{ik}) , \quad (16)$$

where $\delta_{ik} = \delta_i - \delta_k$.

5.2 Newton–Raphson Method

Define mismatch vector

$$\Delta x = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P_{\text{spec}} - P_{\text{calc}} \\ Q_{\text{spec}} - Q_{\text{calc}} \end{bmatrix} . \quad (17)$$

At iteration r ,

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}^{(r)} = \begin{bmatrix} H & N \\ M & L \end{bmatrix}^{(r)} \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}^{(r)} . \quad (18)$$

Update angles and magnitudes until the maximum mismatch is below tolerance.

5.3 Fast Decoupled Power Flow

For high-voltage transmission systems, $X \gg R$, voltage magnitudes are near 1 pu, and angle differences are small. Then active power is strongly coupled to angle and reactive power to voltage:

$$\Delta P / |V| \approx B' \Delta \delta , \quad (19)$$

$$\Delta Q / |V| \approx B'' \Delta |V| . \quad (20)$$

6 Economic Dispatch and Optimal Power Flow

6.1 Economic Dispatch

For generator cost

$$C_i(P_i) = a_i P_i^2 + b_i P_i + c_i , \quad (21)$$

economic dispatch without losses solves

$$\min_{P_i} \sum_i C_i(P_i) \quad \text{s.t.} \quad \sum_i P_i = P_D. \quad (22)$$

The optimality condition is

$$\frac{dC_i}{dP_i} = 2a_i P_i + b_i = \lambda, \quad (23)$$

subject to generator limits.

6.2 AC Optimal Power Flow

AC OPF extends power flow by optimizing an objective subject to network constraints:

$$\min_{V, \delta, P_G, Q_G} f(P_G, Q_G, V) \quad (24)$$

$$\text{s.t.} \quad g(V, \delta, P_G, Q_G) = 0, \quad (25)$$

$$h(V, \delta, P_G, Q_G) \leq 0. \quad (26)$$

Constraints include bus voltage limits, branch thermal limits, generator capability curves, reserve requirements, and security constraints.

7 Short-Circuit and Fault Analysis

7.1 Symmetrical Components

Unbalanced three-phase quantities can be decomposed into positive, negative, and zero sequences:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}, \quad a = e^{j2\pi/3}. \quad (27)$$

The inverse transform is

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}. \quad (28)$$

7.2 Three-Phase Fault

For a bolted balanced three-phase fault at a bus with Thevenin voltage V_{th} and Thevenin impedance Z_1 ,

$$I_f = \frac{V_{th}}{Z_1 + Z_f}. \quad (29)$$

7.3 Single-Line-to-Ground Fault

For an a -phase-to-ground fault:

$$I_a = 3I_0 = \frac{3V_{th}}{Z_1 + Z_2 + Z_0 + 3Z_f}. \quad (30)$$

Grounding and transformer connections strongly affect Z_0 and therefore ground-fault current.

7.4 Fault Current From Synchronous Machines and IBRs

Synchronous generators can contribute large transient currents, often several per-unit initially, limited by subtransient reactance:

$$I_{sc,SG} \approx \frac{E''}{X_d''}. \quad (31)$$

IBRs are current-limited by semiconductors and controls, commonly around 1.1–2.0 pu depending on design and grid code:

$$|\tilde{I}_{IBR}| \leq I_{\max}. \quad (32)$$

This changes protection sensitivity, directionality, relay coordination, and fault-location methods.

8 Protection Fundamentals

Protection detects abnormal conditions and isolates faulted equipment quickly. Common functions include:

- **50/51 overcurrent:** instantaneous/time-delayed current pickup.
- **67 directional overcurrent:** uses voltage polarization to infer fault direction.
- **21 distance:** estimates apparent impedance $Z_{app} = V/I$.
- **87 differential:** compares currents entering and leaving a zone.
- **27/59 undervoltage/overvoltage** and **81U/O frequency** functions.
- **ROCOF:** rate-of-change-of-frequency, df/dt .

IBR Protection Challenge

High IBR penetration reduces short-circuit current magnitude, changes current phase angle through control actions, and can make conventional overcurrent protection less dependable. Protection may need adaptive settings, negative-sequence quantities, differential schemes, transfer trip, traveling waves, or communications-assisted logic.

9 Voltage Control and Reactive Power

9.1 Reactive Power and Voltage

In transmission networks with $X \gg R$, reactive power strongly affects voltage. A useful approximate relation is

$$\Delta V \approx \frac{XQ + RP}{V}. \quad (33)$$

Devices for voltage support include generator automatic voltage regulators (AVRs), shunt capacitors/reactors, static VAR compensators (SVC), STATCOMs, load tap changers, and inverter volt-var controls.

9.2 Generator Capability

Synchronous-generator reactive power is limited by armature current, field current, end-region heating, and stability. IBR reactive capability is limited by current and dc-link constraints:

$$P^2 + Q^2 \leq S_{\text{rated}}^2, \quad I_d^2 + I_q^2 \leq I_{\text{max}}^2. \quad (34)$$

During faults, priority may be given to reactive current injection:

$$I_q^* = k_{\text{FRT}} (V_{\text{ref}} - V_{\text{PCC}}), \quad |I^*| \leq I_{\text{max}}. \quad (35)$$

10 Frequency Control and Inertia

10.1 Swing Equation

The synchronous-machine rotor angle follows

$$M \frac{d^2 \delta}{dt^2} = P_m - P_e - D \frac{d\delta}{dt}, \quad (36)$$

or, in frequency form,

$$\frac{2H}{f_0} \frac{df}{dt} = P_m - P_e - D_f (f - f_0), \quad (37)$$

where H is the inertia constant. Immediately after a generation-loss event, frequency rate of change is approximately

$$\frac{df}{dt} \approx \frac{f_0}{2H_{\text{sys}}} \left(\frac{\Delta P}{S_{\text{base}}} \right). \quad (38)$$

10.2 Primary, Secondary, and Tertiary Control

- **Primary control:** governor or inverter droop responds in seconds.
- **Secondary control:** automatic generation control restores frequency and tie-line interchange in tens of seconds to minutes.
- **Tertiary control:** economic redispatch and reserve management.

11 Stability Concepts

11.1 Rotor-Angle Stability

Rotor-angle stability is the ability of synchronous machines to remain in synchronism. For a classical generator connected to an infinite bus:

$$P_e = \frac{EV}{X} \sin \delta. \quad (39)$$

The synchronizing torque coefficient is

$$K_s = \frac{\partial P_e}{\partial \delta} = \frac{EV}{X} \cos \delta. \quad (40)$$

As δ approaches 90° , synchronizing torque decreases.

11.2 Equal-Area Criterion

For a single-machine infinite-bus system, transient stability can be assessed by comparing accelerating and decelerating areas on the P - δ curve:

$$A_{\text{acc}} = \int_{\delta_0}^{\delta_c} (P_m - P_{e,\text{fault}}) d\delta, \quad A_{\text{dec}} = \int_{\delta_c}^{\delta_{\text{max}}} (P_{e,\text{post}} - P_m) d\delta. \quad (41)$$

Stability requires $A_{\text{dec}} \geq A_{\text{acc}}$.

11.3 Voltage Stability

Voltage stability is the ability to maintain acceptable voltage after a disturbance. A simple radial two-bus example shows the “nose curve”:

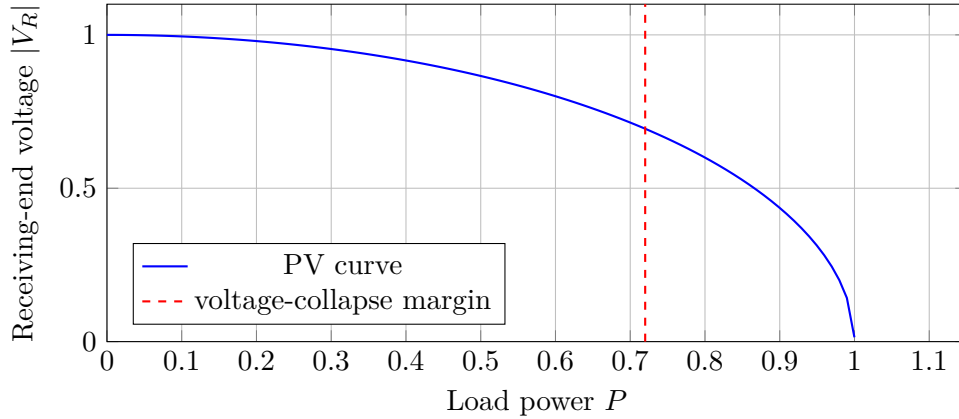


Figure 4: Conceptual PV curve showing maximum loadability and voltage-collapse margin.

11.4 Small-Signal Stability

Linearize nonlinear dynamics $\dot{x} = f(x, u)$ around an operating point:

$$\Delta\dot{x} = A\Delta x + B\Delta u. \quad (42)$$

Eigenvalues $\lambda_i = \sigma_i + j\omega_i$ determine modal behavior. The damping ratio is

$$\zeta_i = \frac{-\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}}. \quad (43)$$

Negative damping or weak inter-area modes can be worsened by controls that interact with grid impedance, PLLs, or other converters.

12 Power Quality

Power quality includes voltage sags, swells, interruptions, harmonics, flicker, unbalance, and transients. Harmonic distortion is quantified by

$$\text{THD}_V = \frac{\sqrt{V_2^2 + V_3^2 + \dots + V_n^2}}{V_1}. \quad (44)$$

Converters use filters and controls to meet harmonic limits. Resonance can occur when inverter filters, cable capacitance, transformer leakage, and grid impedance form lightly damped modes.

13 Inverter-Based Resources

13.1 What Is an IBR?

An IBR interfaces an energy source or storage device through power electronics. Examples include PV inverters, wind turbine full converters, type-3 wind generators with partial converters, BESS, fuel cells, and many HVDC terminals.

13.2 Converter Power Equations in dq Coordinates

With the d -axis aligned to voltage, $v_q = 0$:

$$P = \frac{3}{2}(v_d i_d + v_q i_q) = \frac{3}{2}v_d i_d, \quad (45)$$

$$Q = \frac{3}{2}(v_q i_d - v_d i_q) = -\frac{3}{2}v_d i_q. \quad (46)$$

Thus i_d controls active power and i_q controls reactive power under this axis convention.

13.3 Grid-Following (GFL) Inverters

A GFL inverter behaves like a controlled current source. It synchronizes to an existing grid voltage using a phase-locked loop (PLL), then injects commanded currents.

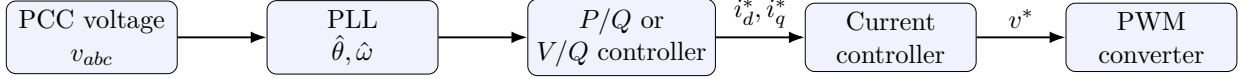


Figure 5: Grid-following inverter control is synchronized by the measured grid voltage.

Typical GFL PLL dynamics include

$$\dot{\hat{\theta}} = \omega_0 + K_p v_q + K_i \int v_q dt. \quad (47)$$

GFL inverters are effective in strong grids but can struggle in weak grids or islanded microgrids because they require a voltage reference to follow.

13.4 Grid-Forming (GFM) Inverters

A GFM inverter behaves more like a controllable voltage source behind an impedance. It establishes voltage magnitude and frequency rather than relying entirely on an external voltage.

Common GFM approaches:

- **Droop control:**

$$\omega = \omega_0 - m_p(P - P_0), \quad (48)$$

$$V = V_0 - n_q(Q - Q_0). \quad (49)$$

- **Virtual synchronous machine (VSM):**

$$2H_v \dot{\omega} = P^* - P - D_v(\omega - \omega_0). \quad (50)$$

- **Dispatchable virtual oscillator control:** nonlinear oscillator dynamics regulate voltage waveform and share power.
- **Matching control:** uses dc-link dynamics to emulate synchronous-machine electromechanical coupling.

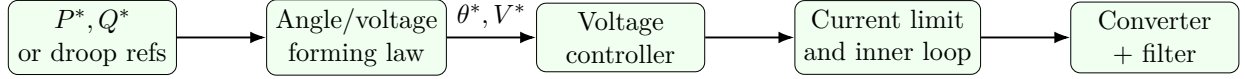


Figure 6: Grid-forming inverter control creates a voltage/frequency reference.

13.5 GFL vs GFM Summary

Feature	Grid-following	Grid-forming
Equivalent behavior	Current source	Voltage source behind impedance
Synchronization	PLL follows grid voltage	Internal oscillator or swing-like law
Islanded operation	Not sufficient alone	Can support islanded microgrid
Weak-grid performance	PLL/control interactions possible	Can improve voltage/frequency stiffness
Fault response	Current-limited injection	Must handle current limiting without losing voltage-forming behavior
Power sharing	Usually outer-loop command	Droop/VSM sharing natural

14 How IBRs Affect Power-Grid Stability

14.1 Reduced Inertia and Faster Dynamics

Replacing synchronous machines with IBRs reduces physical rotating inertia unless the IBR control intentionally provides synthetic inertia or fast frequency response. Lower inertia produces higher RO-COF:

$$\left| \frac{df}{dt} \right| \propto \frac{|\Delta P|}{H_{\text{sys}}}. \quad (51)$$

However, IBRs can respond much faster than mechanical governors if controls, energy headroom, and communications are designed correctly.

14.2 PLL and Weak-Grid Instability

Grid strength is often measured using short-circuit ratio:

$$\text{SCR} = \frac{S_{\text{sc}}}{S_{\text{IBR,rated}}}, \quad S_{\text{sc}} = \frac{V^2}{|Z_{\text{th}}|}. \quad (52)$$

Low SCR means the inverter's own current injection significantly moves the PCC voltage. For GFL controls, PLL dynamics can couple with the weak grid and create oscillations. Effective grid strength can also be reduced by nearby converters interacting through the same network.

14.3 Current Limiting and Protection

IBRs cannot supply unlimited fault current. During severe voltage depressions, the current limit

$$\sqrt{i_d^2 + i_q^2} \leq I_{\text{max}} \quad (53)$$

forces a choice between active current, reactive support, negative-sequence current, and dc-link protection. This affects voltage recovery, relay dependability, and transient stability.

14.4 Control Interactions

High IBR penetration introduces many fast controls: PLLs, current loops, dc-link loops, volt-var controls, plant-level controllers, and grid-forming controllers. Interactions can occur over milliseconds to seconds. Stability studies must include electromagnetic transient (EMT) simulations for fast converter controls when RMS phasor models are insufficient.

14.5 Fault-Induced Delayed Voltage Recovery

IBRs often ride through voltage sags and inject reactive current, but motor loads, feeder controls, and current-limited converters may cause slow voltage recovery. Coordination among DER ride-through, distribution protection, and transmission voltage controls is essential.

15 Microgrids

15.1 Definition and Modes

A microgrid is a local system of loads, DERs, storage, protection, and controls that can operate grid-connected or islanded. It needs:

- at least one grid-forming source for islanded voltage/frequency,
- energy management for power balance,
- protection that works for both grid-connected and islanded fault levels,
- black-start and resynchronization procedures.

15.2 Microgrid Control Hierarchy

1. **Primary control:** local voltage/frequency regulation, droop, current limiting, and virtual impedance.
2. **Secondary control:** restores frequency/voltage deviations caused by droop and coordinates multiple DERs.
3. **Tertiary control:** economic dispatch and power exchange with the main grid.

16 Ride-Through During Faults

16.1 Meaning of Ride-Through

Ride-through means that DERs remain connected and support the grid or microgrid during abnormal voltage/frequency events instead of tripping immediately. Requirements are usually defined by voltage–time and frequency–time envelopes.

16.2 Core Ride-Through Mechanisms

Fault ride-through for an inverter-dominated microgrid usually combines:

1. **Fast fault detection:** voltage sag, current rise, negative/zero sequence, impedance, or differential signals.

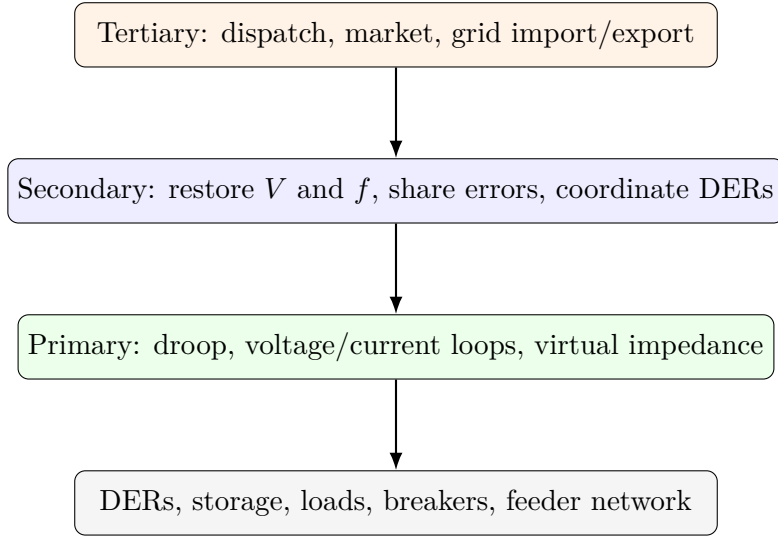


Figure 7: Microgrid hierarchical control.

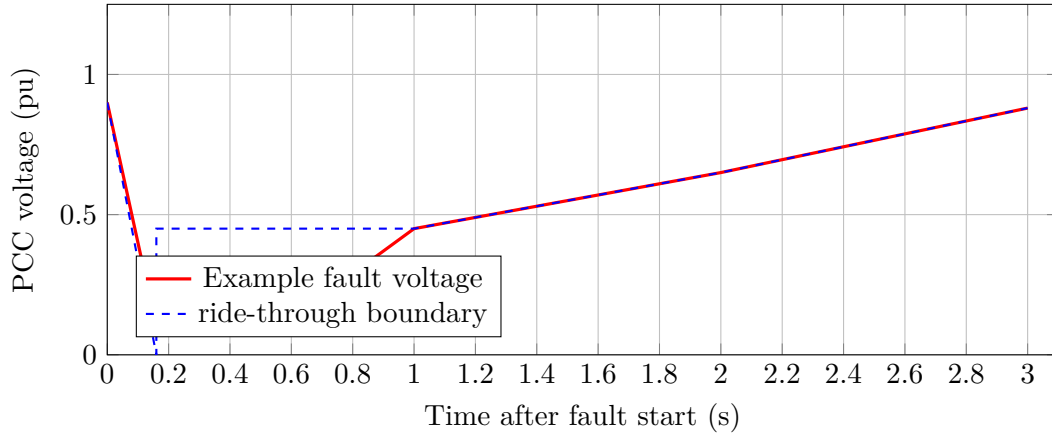


Figure 8: Conceptual low-voltage ride-through envelope. Actual limits depend on grid code.

2. **Current limiting:** protects semiconductors and filters while maintaining a controlled current vector.
3. **Reactive current priority:** supports voltage at the PCC during low-voltage events.
4. **Active power reduction:** prevents dc-link overvoltage when ac power export is limited by sagged voltage.
5. **Energy buffering:** BESS, dc-link capacitance, braking chopper, or curtailment absorbs mismatch.
6. **Protection coordination:** isolates only the faulted zone.
7. **Post-fault recovery:** ramps current, voltage, and power references to avoid secondary dips or converter interactions.

16.3 Current-Limited Control During a Fault

Let $I_{\max}$ be the converter current limit. A common current-priority logic is

$$i_q^* = \text{sat}(k_v(V_{\text{ref}} - V_{\text{PCC}}), -I_{\max}, I_{\max}), \quad (54)$$

$$i_d^* = \text{sgn}(P^*) \sqrt{\max(I_{\max}^2 - (i_q^*)^2, 0)}. \quad (55)$$

If the inverter uses the opposite dq sign convention, the sign of i_q must be changed. For GFM inverters, the current limit must be integrated without abruptly collapsing the voltage reference. Common methods are virtual impedance, voltage reference saturation, current reference limiting, and mode switching to a protected current-source behavior.

16.4 GFM Fault Ride-Through

GFM ride-through is difficult because a voltage source behind low impedance would naturally produce very high fault current. Practical GFM controls add a virtual impedance:

$$v_{\text{cmd}} = v_{\text{gfm}} - R_v i - L_v \frac{di}{dt}, \quad (56)$$

or reduce voltage reference when current approaches the limit:

$$V_{\text{lim}}^* = V^* \min\left(1, \frac{I_{\max}}{|I|}\right). \quad (57)$$

After clearing, the controller restores normal voltage-forming behavior with rate limits:

$$\left| \frac{dP^*}{dt} \right| \leq r_P, \quad \left| \frac{dV^*}{dt} \right| \leq r_V. \quad (58)$$

16.5 Microgrid Fault-Ride-Through Sequence

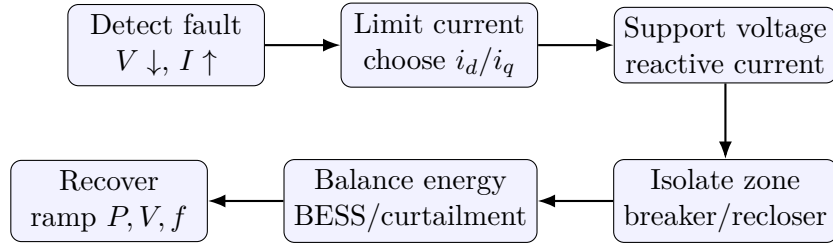


Figure 9: A practical sequence for microgrid fault ride-through.

16.6 Islanded Microgrid Faults

Islanded microgrids have much lower fault current than utility-connected systems, especially when dominated by IBRs. Protection may use:

- differential protection inside critical zones,
- directional elements with adaptive settings,
- voltage-based and sequence-based detection,

- communication-assisted trip,
- temporary overcurrent capability from selected grid-forming BESS,
- fast solid-state breakers or hybrid breakers.

For unbalanced faults, GFM controls may regulate positive sequence while selectively injecting negative-sequence current:

$$\begin{bmatrix} i_1^* \\ i_2^* \end{bmatrix} = \begin{bmatrix} f_1(V_1, P^*, Q^*) \\ f_2(V_2, I_{\max}) \end{bmatrix}, \quad |i_1^* + i_2^*| \leq I_{\max}. \quad (59)$$

17 EMT and DER

17.1 Why EMT Studies Matter for DER-Rich Grids

Electromagnetic transient (EMT) simulation solves instantaneous voltages and currents rather than only positive-sequence phasors. This is important for distributed energy resources (DERs) because converter controls, phase-locked loops, current limiters, dc-link dynamics, filters, and protection logic can act within microseconds to milliseconds. RMS or phasor-domain models are still useful for large-system electromechanical dynamics, but they may hide fast converter instability, unbalanced behavior, harmonics, and protection miscoordination.

When to Use EMT

Use EMT when the study involves inverter controls, weak grids, low short-circuit ratio, unbalanced faults, fast ride-through behavior, harmonics, resonance, sub-cycle protection, islanded microgrids, black start, or detailed switching.

The basic network equations in EMT are differential equations in physical time:

$$v_L(t) = L \frac{di_L(t)}{dt}, \quad (60)$$

$$i_C(t) = C \frac{dv_C(t)}{dt}, \quad (61)$$

$$v_R(t) = Ri_R(t). \quad (62)$$

Numerical integration converts these equations into discrete-time companion models. If the EMT time step is Δt , it must be small enough to represent the fastest control, switching, and network modes of interest.

17.2 DER Types and EMT Modeling Blocks

DERs include rooftop PV, utility-scale PV connected at distribution voltages, BESS, wind, fuel cells, microturbines, electric vehicles, and controllable loads. A detailed EMT DER model commonly contains:

- dc source or machine-side model,
- dc-link capacitor and protection,
- voltage-source converter bridge,
- pulse-width modulation or averaged converter model,
- L , LC , or LCL filter,

- transformer and grounding model,
- PLL or grid-forming angle generator,
- inner current/voltage loops,
- outer active/reactive power, volt-var, volt-watt, and frequency-watt loops,
- current limiting and ride-through state machines,
- plant controller and communication delays where relevant.

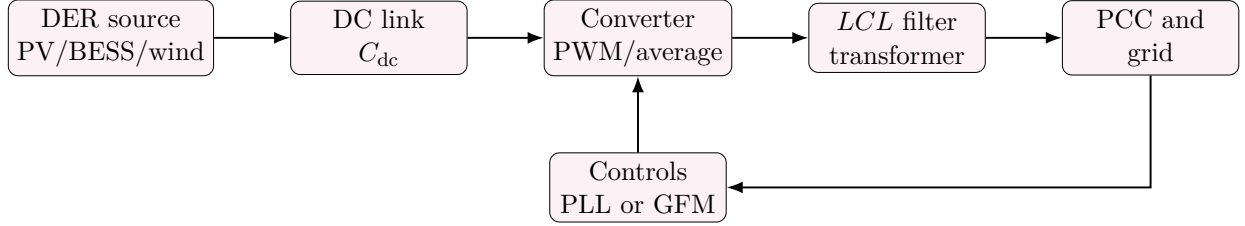


Figure 10: Typical EMT modeling blocks for an inverter-interfaced DER.

17.3 Faults and Short-Circuit Currents in DER Systems

Classical short-circuit analysis assumes fault current is driven by voltage sources behind impedances. This is often valid for synchronous machines:

$$I_{\text{fault,SG}} \approx \frac{E''}{R_s + jX_d'' + Z_f}. \quad (63)$$

DER inverters behave differently because semiconductor ratings and controls limit current:

$$|\tilde{I}_{\text{fault,DER}}| \leq I_{\text{max}}. \quad (64)$$

The current waveform may be controlled to prioritize reactive current, positive sequence, negative sequence, dc-link protection, or thermal protection. Therefore fault current magnitude, phase angle, and sequence content are control-dependent.

Protection implication. In a DER-rich feeder, the fault current seen by an overcurrent relay can be lower in islanded mode than in grid-connected mode:

$$I_{\text{fault,grid connected}} \gg I_{\text{fault,islanded}}. \quad (65)$$

This motivates adaptive settings, differential protection, directional elements, transfer trip, communication-assisted schemes, or voltage/sequence logic.

Unbalanced faults. For line-to-ground and line-to-line faults, DER controls may suppress or command negative-sequence current. A current-limited positive/negative-sequence command must satisfy

$$|i_1^* + i_2^* + i_0^*| \leq I_{\text{max}}. \quad (66)$$

If a model assumes no negative-sequence current when the real inverter injects some, relay behavior and voltage unbalance predictions may be wrong.

17.4 Weak Grids and Short-Circuit Ratio

Grid strength describes how much the PCC voltage changes when the DER injects current. A common metric is short-circuit ratio:

$$\text{SCR} = \frac{S_{\text{sc}}}{S_{\text{DER,rated}}} = \frac{V_{\text{PCC}}^2 / |Z_{\text{th}}|}{S_{\text{DER,rated}}}. \quad (67)$$

Lower SCR means a weaker grid. In weak grids, the inverter is not merely responding to the grid voltage; its own current injection significantly changes that voltage:

$$\Delta V_{\text{PCC}} \approx Z_{\text{th}} \Delta I_{\text{DER}}. \quad (68)$$

SCR range	Typical interpretation
$\text{SCR} > 5$	Strong grid for many conventional GFL controls
$3 < \text{SCR} \leq 5$	Moderate grid; controls should be checked
$2 < \text{SCR} \leq 3$	Weak grid; EMT and control tuning often needed
$\text{SCR} \leq 2$	Very weak grid; GFM or special controls may be required

For multiple nearby DER plants, weighted short-circuit ratio (WSCR) or impedance-based methods may be more informative than individual SCR, because plants interact through the shared network.

17.5 Low Inertia and Fast DER Frequency Response

Synchronous-machine inertia naturally resists frequency changes. Inverter-based DERs do not inherently provide mechanical inertia, so high DER penetration can increase ROCOF after a disturbance:

$$\frac{df}{dt} \approx \frac{f_0}{2H_{\text{sys}}} \frac{\Delta P}{S_{\text{base}}}. \quad (69)$$

DERs can still support frequency through fast frequency response:

$$\Delta P_{\text{droop}} = -K_f(f - f_0), \quad (70)$$

$$P_{\text{synthetic inertia}} = -K_{\text{in}} \frac{df}{dt}. \quad (71)$$

These controls require available headroom and energy. PV may need curtailment to increase output after underfrequency. BESS can inject or absorb power rapidly, but it is limited by state of charge, current rating, and thermal constraints.

Low-Inertia Design Issue

Fast DER response improves frequency nadir only if measurement delay, control delay, communication delay, and power limits are included. A response that is too aggressive can excite oscillations in weak grids or microgrids.

17.6 PLL Dynamics and Stability

A grid-following DER uses a PLL to estimate the grid voltage angle. In the synchronous reference frame, a simple PLL drives v_q toward zero:

$$\dot{\xi}_{\text{PLL}} = v_q, \quad (72)$$

$$\hat{\omega} = \omega_0 + K_p v_q + K_i \xi_{\text{PLL}}, \quad (73)$$

$$\dot{\hat{\theta}} = \hat{\omega}. \quad (74)$$

In a strong grid, the PCC voltage is nearly unaffected by inverter current, so the PLL sees a stiff reference. In a weak grid, inverter current changes the PCC voltage angle and magnitude; the PLL then interacts with the current controller and grid impedance. This can cause low-frequency oscillations, loss of synchronism, or false ride-through behavior.

PLL tuning tradeoff. A fast PLL tracks angle jumps quickly but is more sensitive to voltage distortion, unbalance, and weak-grid coupling. A slow PLL filters disturbances better but may track faults and phase jumps poorly. EMT studies should test faults, voltage phase-angle jumps, weak-grid dispatch points, and neighboring converter interactions.

17.7 IEEE 1547 and DER Interconnection Behavior

IEEE 1547 defines interconnection and interoperability requirements for DERs connected to electric power systems. Exact settings depend on the adopted edition, authority having jurisdiction, utility requirements, DER category, and application. Important study concepts include:

- voltage and frequency ride-through requirements,
- mandatory voltage and frequency trip settings,
- reactive power capability and voltage regulation modes,
- volt-var, volt-watt, frequency-watt, and constant power factor functions,
- response to abnormal voltage and frequency,
- interoperability and communications,
- limitations on unintentional islanding,
- power quality requirements such as harmonics and flicker.

Common DER grid-support functions. Volt-var control adjusts reactive power based on PCC voltage:

$$Q^* = f_{VV}(V_{PCC}). \quad (75)$$

Volt-watt control reduces active power at high voltage:

$$P^* = f_{VW}(V_{PCC}). \quad (76)$$

Frequency-watt control changes active power with frequency:

$$P^* = P_0 - K_{FW}(f - f_0). \quad (77)$$

These functions can improve grid support, but many DERs responding with similar curves can interact with feeder voltage regulators, capacitor banks, and each other.

17.8 DER Ride-Through in EMT

During a fault, DER ride-through can be represented as a sequence of control states:

1. detect abnormal voltage, frequency, current, or sequence component,
2. enter ride-through mode if the event is inside the required envelope,

3. limit current while prioritizing required support,
4. manage dc-link energy through curtailment, BESS, or chopper,
5. avoid nuisance tripping while protecting the converter,
6. ramp back to normal commands after voltage and frequency recover.

The EMT model should include saturation and limiters because they dominate behavior during faults:

$$i_q^* = \text{sat}(k_v(V_{\text{ref}} - V_{\text{PCC}}), -I_{\text{max}}, I_{\text{max}}), \quad (78)$$

$$i_d^* = \text{sat}\left(\frac{2P^*}{3v_d}, -I_{d,\text{max}}, I_{d,\text{max}}\right), \quad (79)$$

$$\sqrt{(i_d^*)^2 + (i_q^*)^2} \leq I_{\text{max}}. \quad (80)$$

If the grid code gives reactive current priority, i_q^* is assigned first and the remaining current headroom is used for i_d^* .

17.9 EMT Study Checklist for DER Interconnection

An EMT DER study should include:

- validated DER models for all relevant operating modes,
- short-circuit ratio and grid impedance sensitivity,
- balanced and unbalanced faults at the PCC and nearby buses,
- fault clearing, reclosing, and voltage phase-angle jumps,
- IEEE 1547 ride-through and trip settings used by the project,
- PLL and current-control bandwidth sensitivity,
- harmonic and resonance scans for filters, cables, and capacitor banks,
- islanding, black-start, and resynchronization cases for microgrids,
- protection dependability and security in grid-connected and islanded modes,
- post-fault active-power recovery and voltage recovery performance.

17.10 Related Advanced Topics

Impedance-based stability. Converter-grid oscillations can be assessed by measuring or modeling converter output impedance and grid impedance. Instability risk increases when impedance ratios encircle critical points in a Nyquist test:

$$L(s) = Z_{\text{grid}}(s)Y_{\text{converter}}(s). \quad (81)$$

Harmonics and resonance. An *LCL* filter resonance frequency is approximately

$$\omega_r = \sqrt{\frac{L_1 + L_2}{L_1 L_2 C_f}}. \quad (82)$$

Weak grids shift damping and resonance behavior because the grid impedance is part of the filter seen by the converter.

EMT model fidelity. Switching models capture semiconductor switching and harmonics but require small time steps. Averaged models are faster and often sufficient for control and ride-through studies if harmonics and switching transients are not the focus. The model should be no more detailed than necessary, but detailed enough to represent the phenomenon being studied.

18 Advanced Study Methods

18.1 RMS Phasor vs EMT Simulation

- **RMS/transient-stability simulation:** efficient for electromechanical dynamics, governor/AVR response, and large systems.
- **EMT simulation:** necessary for converter switching, PLL dynamics, unbalanced faults, harmonics, sub-synchronous/control interactions, and protection details.

High IBR systems often require EMT model validation because average phasor models may hide fast control instabilities.

18.2 Impedance-Based Stability

Converter-grid interactions can be studied using source and load impedances. A small-signal stability condition can be framed with the minor-loop gain

$$L(s) = Z_{\text{source}}(s)Y_{\text{load}}(s), \quad (83)$$

and Nyquist criteria. Weak grids have larger Z_{source} , making interaction with converter input admittance more likely.

18.3 Resonance and Sub-Synchronous Oscillation

Series compensation, wind plants, HVDC, and converter controls can create oscillations below or above the fundamental frequency. Modal damping, impedance scans, and EMT studies are used to identify risk.

19 Extended Topic Details and Worked Examples

19.1 Per-Unit: Practical Workflow and Example

The most common per-unit mistakes are mixing line-to-line and phase voltage bases, forgetting to change base after a transformer, and using single-phase power with a three-phase voltage base. A safe workflow is:

1. Select a common three-phase power base S_{base} for the study.
2. Select a line-to-line voltage base V_{base} in one zone.
3. Propagate voltage bases through ideal transformer ratios.
4. Convert all impedances, admittances, voltages, currents, and powers to per-unit before solving.
5. Convert final answers back to physical units only after the network solution is complete.

Example. A transformer leakage reactance is $X = 8\%$ on 50 MVA, 13.8 kV. On a 100 MVA, 13.8 kV base,

$$X_{\text{new}} = 0.08 \left(\frac{100}{50} \right) \left(\frac{13.8}{13.8} \right)^2 = 0.16 \text{ pu.} \quad (84)$$

If the voltage base also changed, the voltage ratio factor would be squared.

Per-Unit Study Tip

For transformer-connected networks, choose voltage bases consistent with nominal transformer ratios. Then ideal transformers disappear from the per-unit circuit and only leakage impedances remain.

19.2 Detailed Bus Modeling

Bus type is not just a mathematical classification; it represents a physical control behavior.

- A **slack bus** represents a source with enough flexibility to absorb modeling mismatch, losses, and the missing system angle reference. In large interconnections, the real imbalance is physically shared by many governors, but one slack is used numerically.
- A **PV bus** represents a voltage-regulated generator, STATCOM, or inverter with a voltage-control mode. The controller adjusts Q to keep $|V|$ at the scheduled value.
- A **PQ bus** represents a fixed complex-power injection or load. Constant-power loads are nonlinear because current increases as voltage decreases:

$$I^* = \frac{S}{V}. \quad (85)$$

Load models often combine constant impedance, current, and power terms:

$$P(V) = P_0 \left(a_Z \left(\frac{V}{V_0} \right)^2 a_I \left(\frac{V}{V_0} \right) + a_P \right), \quad (86)$$

$$Q(V) = Q_0 \left(b_Z \left(\frac{V}{V_0} \right)^2 b_I \left(\frac{V}{V_0} \right) + b_P \right). \quad (87)$$

This is called a ZIP load model.

19.3 Power-Flow Details Beyond the Equations

Mismatch interpretation. At every non-slack bus, power-flow mismatch is the difference between specified net injection and calculated network injection:

$$\Delta S_i = S_{i,\text{spec}} - V_i \left(\sum_k Y_{ik} V_k \right)^*. \quad (88)$$

A good solution has small mismatches and all limits respected.

PV-to-PQ switching. If a generator reaches its reactive limits,

$$Q_{\min} \leq Q_G \leq Q_{\max}, \quad (89)$$

it can no longer regulate voltage. The bus is converted from PV to PQ with Q_G fixed at the violated limit. This often causes nearby voltages to drop and can reveal voltage-stability weakness.

Continuation power flow. For voltage-stability studies, loading is increased by a parameter λ :

$$P_L = P_{L0}(1 + \lambda), \quad Q_L = Q_{L0}(1 + \lambda). \quad (90)$$

Continuation methods trace the solution through the nose point where the ordinary Newton Jacobian becomes singular:

$$\det(J) \rightarrow 0. \quad (91)$$

19.4 Transmission and Distribution Modeling Details

Transmission grids are usually inductive, meshed, and high voltage. Distribution systems are often radial, have larger R/X ratios, and may be unbalanced. Therefore, distribution power flow may need phase-domain equations:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} Y_{aa} & Y_{ab} & Y_{ac} \\ Y_{ba} & Y_{bb} & Y_{bc} \\ Y_{ca} & Y_{cb} & Y_{cc} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}. \quad (92)$$

Untransposed lines, single-phase laterals, rooftop PV, and unequal loads can make positive-sequence approximations inaccurate.

19.5 Fault Analysis: More Fault Types

Besides three-phase and single-line-to-ground faults, common unbalanced faults include line-to-line (LL) and double-line-to-ground (LLG).

Line-to-line fault. For a bolted b - c fault,

$$I_1 = \frac{V_{th}}{Z_1 + Z_2 + Z_f}, \quad I_0 = 0, \quad I_2 = -I_1. \quad (93)$$

Double-line-to-ground fault. For a b - c -ground fault, the sequence networks are connected in a series-parallel arrangement. With fault impedance omitted for clarity,

$$I_1 = \frac{V_{th}}{Z_1 + (Z_2 \parallel Z_0)}. \quad (94)$$

LLG faults can create large zero-sequence currents depending on grounding.

DC offset and breaker duty. Fault current is not purely sinusoidal immediately after inception. A decaying dc component depends on the fault point-on-wave and X/R ratio:

$$i(t) = I_{ac} \sin(\omega t + \phi) + I_{dc,0} e^{-t/\tau}, \quad \tau = \frac{L}{R}. \quad (95)$$

High X/R systems have slower dc decay and higher asymmetrical breaker duty.

19.6 Protection Coordination Details

Protection must satisfy sensitivity, selectivity, speed, security, and dependability. These goals conflict. Faster settings may reduce equipment damage but increase misoperation risk.

Inverse-time overcurrent. A generic inverse-time relay curve can be expressed as

$$t = \text{TMS} \frac{A}{(I/I_{\text{pickup}})^p - 1} + B, \quad (96)$$

where TMS is the time multiplier setting. Coordination requires an upstream device to operate after the downstream device by a coordination time interval.

Distance relay zones. Distance relays typically use:

- Zone 1: instantaneous underreaching protection, often 80–90% of line.
- Zone 2: time-delayed overreach into the next line.
- Zone 3: remote backup with longer delay.

IBR controls can distort apparent impedance because current magnitude and angle are actively controlled rather than governed by a passive source impedance.

19.7 Voltage Control: Local and System-Level Details

Voltage control is hierarchical. Local devices act quickly, while operators and energy-management systems coordinate schedules over minutes.

$$Q_{\text{cap}} = \omega C V^2, \quad Q_{\text{reactor}} = -\frac{V^2}{\omega L}. \quad (97)$$

Capacitors provide more reactive power at high voltage and less at low voltage, while STATCOMs can provide nearly constant current support over a wider voltage range:

$$Q_{\text{STATCOM}} \approx V I_q. \quad (98)$$

Thus STATCOM reactive power falls linearly with voltage, while capacitor VARs fall approximately with V^2 .

19.8 Frequency Control With IBRs

In low-inertia systems, frequency nadir is affected by inertia, response delay, droop gain, available headroom, and energy limits. A simple droop law is

$$\Delta P = -\frac{1}{R} \Delta f, \quad (99)$$

where R is the droop coefficient. Synthetic inertia from an inverter may use

$$P_{\text{inertia}} = -K_{\text{in}} \frac{df}{dt}, \quad (100)$$

but it requires measurement filtering because differentiating frequency amplifies noise. BESS can provide this response naturally if state of charge, thermal limits, and converter ratings allow it.

19.9 Stability: Time Scales and Study Models

Power-system stability is best understood by time scale:

Phenomenon	Typical time scale	Common model
Converter switching and controls	microseconds–milliseconds	EMT
Fault current and protection	milliseconds–cycles	EMT or phasor
Rotor-angle transient stability	cycles–seconds	transient stability
Primary frequency response	seconds–tens of seconds	dynamic phasor/RMS
Voltage recovery and load dynamics	seconds–minutes	RMS plus load models
AGC and dispatch	minutes	steady-state/dynamic optimization

Transient stability margin. Critical clearing time (CCT) is the maximum fault duration that preserves stability. Reducing transfer reactance, increasing voltage support, faster clearing, and fast active-power controls can improve CCT:

$$t_{\text{clear}} < t_{\text{critical}}. \quad (101)$$

Voltage stability indicators. Common indicators include reactive reserve, PV curve margin, QV curve margin, Jacobian singular values, and Thevenin equivalent impedance ratio:

$$\left| \frac{Z_{\text{th}}}{Z_{\text{load}}} \right| \rightarrow 1 \quad \text{near maximum power transfer.} \quad (102)$$

19.10 IBR Modeling Details

An IBR plant model usually includes:

- energy source model, such as PV array or wind turbine aerodynamics,
- dc-link dynamics,
- converter current and voltage limits,
- inner current loop,
- PLL or grid-forming oscillator,
- plant-level controller,
- transformer, collector system, and cable models,
- protection and ride-through logic.

The dc-link energy balance is

$$C_{\text{dc}} V_{\text{dc}} \frac{dV_{\text{dc}}}{dt} = P_{\text{dc}} - P_{\text{ac}} - P_{\text{loss}}. \quad (103)$$

During a voltage sag, P_{ac} may fall because

$$P_{\text{ac}} \leq \frac{3}{2} V_{\text{PCC}} I_{\text{max}}. \quad (104)$$

If P_{dc} is not curtailed or stored, V_{dc} rises.

19.11 GFL Control Details

A typical GFL control stack contains:

1. PLL estimates angle and frequency.
2. Outer loops compute i_d^* and i_q^* from P/Q , V/Q , or dc-link objectives.
3. Current limiter enforces converter capability.
4. Inner current controller generates voltage commands.
5. PWM or modulation applies switching signals.

With an L filter, a simplified dq current controller has plant dynamics

$$L \frac{di_d}{dt} = v_{cd} - v_{gd} + \omega L i_q - R i_d, \quad (105)$$

$$L \frac{di_q}{dt} = v_{cq} - v_{gq} - \omega L i_d - R i_q. \quad (106)$$

Feedforward decoupling terms $\pm \omega L i$ improve tracking but depend on accurate frequency estimation.

19.12 GFM Control Details

GFM inverters must regulate voltage while sharing load with other sources. Droop control gives decentralized sharing:

$$\Delta P_i \approx \frac{1/m_i}{\sum_k 1/m_k} \Delta P_{\text{load}}. \quad (107)$$

Smaller droop slope m_i means stiffer frequency behavior and a larger share of load changes. Virtual impedance improves sharing and limits circulating current:

$$v_{\text{terminal}}^* = v_{\text{internal}}^* - Z_v i. \quad (108)$$

The tradeoff is voltage regulation: larger virtual impedance improves stability and sharing but increases voltage drop under load.

19.13 High-IBR Stability Challenges

High IBR penetration changes grid behavior in several ways:

- **Lower mechanical inertia:** higher ROCOF after contingencies.
- **Lower fault current:** reduced overcurrent relay sensitivity.
- **Fast controls:** possible control-loop interactions and oscillations.
- **Weak-grid operation:** inverter terminal voltage is strongly affected by inverter current.
- **Model uncertainty:** vendor controls may be proprietary, so validated black-box or user-defined models are needed.
- **Resource tripping risk:** poorly coordinated ride-through can turn a local disturbance into a large generation-loss event.

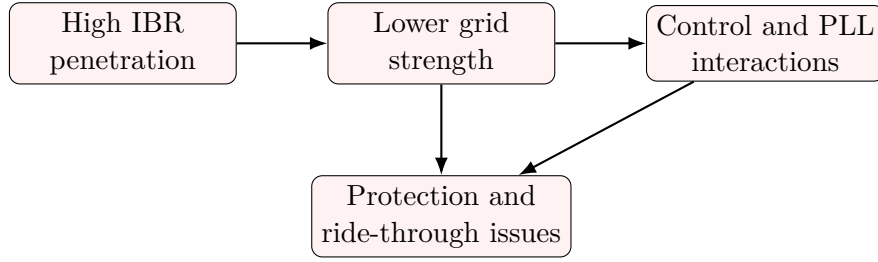


Figure 11: Main stability and protection concerns as IBR penetration increases.

19.14 Microgrid Design Details

A practical microgrid design should answer these questions:

1. Which DER forms voltage and frequency when islanded?
2. Is there enough spinning or synthetic reserve for the largest load step?
3. Can the energy storage sustain islanded operation for the required time?
4. Do protective devices detect faults in both grid-connected and islanded modes?
5. How is the point of common coupling opened and resynchronized?
6. Are black-start sequences defined and tested?

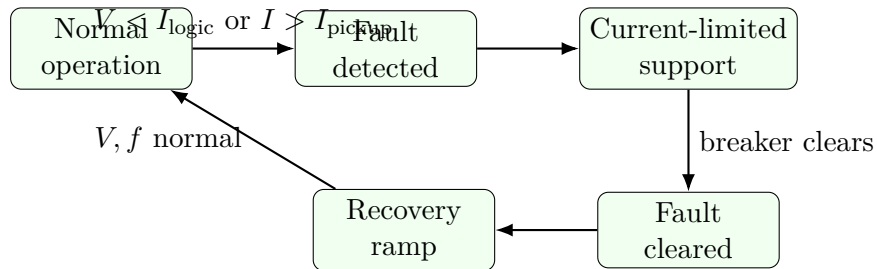
For islanded power balance,

$$\sum P_{\text{DER}} - \sum P_{\text{load}} - P_{\text{loss}} = 2H_{\text{eq}}\omega \frac{d\omega}{dt}. \quad (109)$$

In an inverter-only microgrid, H_{eq} is a control parameter rather than physical inertia, and energy must still come from dc sources or storage.

19.15 Fault Ride-Through: Detailed Control Logic

The ride-through controller can be represented as a state machine:



In each state, limits must be explicit:

$$|I| \leq I_{\text{max}}, \quad (110)$$

$$V_{\text{dc,min}} \leq V_{\text{dc}} \leq V_{\text{dc,max}}, \quad (111)$$

$$T_{\text{device}} \leq T_{\text{max}}, \quad (112)$$

$$\left| \frac{di^*}{dt} \right| \leq r_i. \quad (113)$$

Why ride-through is hard in a microgrid. The same inverter may be responsible for voltage formation, fault current, frequency regulation, and energy balancing. During a severe fault, these goals conflict. A useful priority order is:

device protection > safety/protection coordination > voltage support > active-power delivery. (114)

19.16 Model Validation and Study Checklist

For real projects, a study should document:

- base cases, contingencies, and operating conditions,
- model source and validation status,
- protection settings and breaker clearing times,
- inverter current limits and control modes,
- voltage/frequency ride-through curves,
- load composition, especially motors and power-electronic loads,
- grid strength metrics at each interconnection point,
- sensitivity studies for dispatch, topology, and controller gains.

20 Common Exam and Interview Questions

1. Why is one bus chosen as slack in power flow?
2. What happens when a PV bus reaches reactive power limits?
3. Why does P mostly depend on angle and Q mostly on voltage in a transmission grid?
4. How do IBR fault currents differ from synchronous-generator fault currents?
5. Why can a PLL destabilize a GFL inverter in a weak grid?
6. What makes a GFM inverter useful for an islanded microgrid?
7. How is low-voltage ride-through achieved without damaging the converter?
8. Why can high IBR penetration require EMT instead of only RMS simulation?
9. How do virtual impedance and current limiting help GFM fault ride-through?
10. What protection functions remain dependable when fault current is low?

21 Quick Reference

Essential Equations

$$\begin{aligned}
 S &= VI^* = P + jQ \\
 Z_{\text{base}} &= V_{\text{base}}^2 / S_{\text{base}} \\
 I &= Y_{\text{bus}} V \\
 P_i &= |V_i| \sum_k |V_k| (G_{ik} \cos \delta_{ik} + B_{ik} \sin \delta_{ik}) \\
 Q_i &= |V_i| \sum_k |V_k| (G_{ik} \sin \delta_{ik} - B_{ik} \cos \delta_{ik})
 \end{aligned} \tag{28}$$

Study Priorities

- Know bus types and power-flow equations.
- Understand voltage, angle, and frequency stability separately.
- Compare SG and IBR fault behavior.
- Distinguish GFL current-source behavior from GFM voltage-source behavior.
- For microgrids, connect controls, energy balance, and protection.

22 Suggested Learning Path

1. Review phasors, complex power, and per-unit calculations.
2. Build Y_{bus} for a three-bus system and solve power flow.
3. Study symmetrical components and compute SLG and three-phase faults.
4. Simulate swing-equation response to a generation trip.
5. Compare GFL and GFM inverter block diagrams and control equations.
6. Analyze how current limits alter fault response and voltage support.
7. Study a microgrid fault sequence: detect, limit, support, isolate, recover.
8. Move from RMS to EMT simulation when converter controls dominate dynamics.